

Assignment 1

14X060 Quantum information theory

Exercise 11.3

Prove that the binary entropy $H_{\text{bin}}(p)$ attains its maximum value of one at $p = 1/2$

Exercise 11.5: Sub-additivity of the Shannon entropy

Show that $H(p(x, y) \| p(x)p(y)) = H(p(x)) + H(p(y)) - H(p(x, y))$. Deduce that $H(X, Y) \leq H(X) + H(Y)$, with equality if and only if X and Y are independent random variables.

Exercise 11.15: Generalised measurements can decrease entropy

Suppose a qubit is in the state ρ is measured using the measurement operators $M_1 = |0\rangle\langle 0|$ and $M_2 = |0\rangle\langle 1|$. If the result of the measurement is unknown to us then the state of the system afterwards is $M_1\rho M_1^\dagger + M_2\rho M_2^\dagger$. Show that this procedure can *decrease* the entropy of the qubit.

Exercise 12.1

Suppose $|\psi\rangle$ and $|\varphi\rangle$ are two orthogonal quantum states of a single qubit. Design a quantum circuit with two input qubits (the 'data' and the 'target' qubits), with the data qubit in either the state $|\psi\rangle$ or $|\varphi\rangle$, and the target qubit prepared in the standard state $|0\rangle$, which produces as output $|\psi\rangle|\psi\rangle$ or $|\varphi\rangle|\varphi\rangle$, depending on whether $|\psi\rangle$ or $|\varphi\rangle$ was input to the data qubit.

Exercise 12.4

Suppose Alice sends Bob an equal mixture of the four pure states

$$\begin{aligned} |X_1\rangle &= |0\rangle \\ |X_2\rangle &= \sqrt{\frac{1}{3}} [|0\rangle + \sqrt{2}|1\rangle] \\ |X_3\rangle &= \sqrt{\frac{1}{3}} [|0\rangle + \sqrt{2}e^{2\pi i/3}|1\rangle] \\ |X_4\rangle &= \sqrt{\frac{1}{3}} [|0\rangle + \sqrt{2}e^{4\pi i/3}|1\rangle] \end{aligned}$$

Show that the maximum mutual information between Bob's measurement and Alice's transmission is less than one bit. A POVM which achieves ≈ 0.415 bits is known. Can you construct this or better yet, one which achieves the Holevo bound?

To archive all results (solutions, report together with your code, etc.) and submit them using Quantum computing course moodle.